

Enhancing Cybersecurity for LDACS: a Secure and Lightweight Mutual Authentication and Key Agreement Protocol

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Abstract—The aviation industry faces significant challenges due to rising global air travel demand. Frequency saturation in Air Traffic Management (ATM) leads to communication problems, necessitating the enhancement of traditional systems. The Single European Sky ATM Research (SESAR) initiative, backed by the European Commission, aims to digitize ATM, with the L-band Digital Aeronautical Communications System (LDACS) as a key component. LDACS aims to improve communication, enhance surveillance, and optimize airspace usage for safer, more efficient ATM. Although LDACS is protected against most cyberattacks, a critical security objective, anonymity, is currently overlooked. To strengthen LDACS's security, robust authentication mechanisms, Post-Quantum security, and measures to ensure aircraft anonymity are crucial. Therefore, we propose a comprehensive security framework to enhance LDACS's cybersecurity, focusing on mutual authentication and key agreement. The protocol uses Physical Unclonable Function (PUF) for robust mutual authentication and Bit-flipping Key Encapsulation (BIKE) for secure session key establishment utilizing Post-Quantum Cryptography (PQC). This framework ensures anonymity and secure communication between aircraft and ground stations while minimizing message exchange, latency, and data overhead. An informal security analysis confirms our proposed framework's potential to augment the efficiency and security of ATM operations.

Index Terms—Aviation Cybersecurity, BIKE, LDACS, PUF, Mutual Authentication and Key Exchange (MAKE).

I. INTRODUCTION

Air traffic control (ATC) plays a vital role in air transportation's safe and efficient operation, especially at busy airports like London Heathrow, which often handles over 1500 take-offs and landings daily. The growing number of

flights has created challenges for ATC, requiring them to manage and coordinate a high volume of aircraft movements to ensure safety and minimize delays. Experts predict that worldwide flight movements could double by 2030, putting more pressure on the existing ATC infrastructure [1]. Traditional aviation communication systems, such as Very High Frequency (VHF), face challenges in managing the growing air traffic. These analog systems encounter frequency congestion problems caused by the high volume of flights and inefficient spectrum utilization, resulting in unreliable communication.

In response to the increasing burden on ATM, the aviation industry has recognized the need to improve traditional communication systems [2]. The current VHF systems, which are reaching their limits, particularly in high-density areas, are unable to cope with the growing air traffic. Consequently, the Federal Aviation Administration (FAA) and the European Organisation for the Safety of Air Navigation (EUROCONTROL) collaborated on Action Plan 17 [3], which confirmed the inadequacy of existing technology for future requirements.

In response to these challenges, digital solutions such as Controller-Pilot Data Link Communications (CPDLC) [4], [5] and Automatic Dependent Surveillance-Broadcast (ADS-B) [6] have been implemented and are currently in use. These technologies strive to enhance communication, improve surveillance capabilities, and optimize airspace utilization for safer and more efficient air traffic management. Europe's SESAR program [7], and the United States' Next Generation National Airspace System (NextGen) project [8] are notable initiatives leading these efforts. Despite the significant improvements brought about by CPDLC and ADS-B, these systems do have limitations. They present cybersecurity vulnerabilities, lack secure communication methods, and face

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scalability issues in light of increasing air traffic volume. These challenges motivate the development of the Future Communication Infrastructure (FCI) and, in turn, the LDACS.

The FCI is a collaborative development by SESAR and NextGen, aiming to revolutionize ATC communication capabilities by introducing new datalinks and transitioning to an IP-based aeronautical telecommunications network. The FCI focuses on enhancing the capacity and performance of VHF systems by implementing standardized broadband digital data link technologies. The FCI optimizes network management and improves data transmission efficiency by incorporating digital data links, adaptive modulation, and coding techniques [9]. One significant advancement within the FCI is LDACS [10]. LDACS operates in the L-band and offers substantial improvements in bandwidth, capacity, and coverage compared to traditional VHF and CPDLC systems [11]. It provides an innovative solution to address the limitations of existing communication systems in the aviation industry.

Furthermore, LDACS supports robust security features to ensure aeronautical communications confidentiality, integrity, and availability [12], [13]. As a critical component of the FCI, it incorporates encryption mechanisms, authentication protocols, and other protective measures. These features are designed to protect against unauthorized access and potential cyber threats. However, LDACS is still in the process of standardizing protocols for mutual authentication and key establishment between the aircraft and ATC. While research is actively underway, no definitive standards have been established yet. This critical aspect necessitates ongoing attention to identifying the most suitable security measures, ensuring enhanced protection for LDACS communications.

To enhance security in LDACS, researchers [12]–[15] have proposed different cybersecurity solutions for LDACS. These solutions aim to implement robust mutual authentication mechanisms to ensure secure communication between aircraft and ground stations. However, integrating these security measures into real-time systems presents challenges, including increased data overhead and authentication latency. Striking a balance between robust security and minimal impact on system performance is crucial. Therefore, developing lightweight security solutions that effectively address vulnerabilities while meeting LDACS's real-time requirements becomes necessary.

Therefore, this paper introduces a lightweight framework designed specifically for LDACS. The framework aims to minimize data overhead and authentication latency by reducing message exchanges, enabling faster and more efficient communication while maintaining strong security measures. Similar to the works in [13], the integration of PQC, such as the BIKE [16] protocol, enhances long-term security, provides future-proofing against potential quantum threats, ensures compliance with the aviation industry standards, and reinforces trust and confidence in LDACS.

A. Our Contributions

- We propose a lightweight, robust security protocol designed explicitly for securing air-ground communica-

tion in LDACS. It provides mutual authentication and key establishment, leveraging lightweight cryptography primitives such as Message Authentication Code (MAC), nonce, PUF, and bitwise XOR operations.

- We integrate the BIKE protocol into our proposed scheme, enabling the generation of a robust, secure session key. This integration ensures the system's resilience against post-quantum cyber attacks, safeguarding the long-term security of LDACS.
- We propose a solution that substantially contributes to LDACS by reducing communication overhead and computation costs and improving system performance.
- We verify the robustness of the proposed solution through informal analysis, ensuring its resilience against various malicious attacks.

B. Paper Organization

This paper is structured as follows: Section II delivers a comprehensive background overview and the relevant literature on LDACS. Section III deals with the adversary model and security goals. Section IV presents the essential preliminaries for our protocol. Section V delves into the proposed mutual authentication and key exchange, followed by an informal security analysis in Section VI. Section VII discusses the performance of the proposed solution. Finally, Section VIII concludes the paper and discusses future work.

II. BACKGROUND AND RELATED WORK

This section explains the background and communication components of LDACS, as well as an overview of existing related work in the field of LDACS.

A. LDACS Background

The L-band Digital Aeronautical Communications System (LDACS) is a novel air-ground communication system currently under development in Europe. It aims to provide secure and efficient communication between Aircraft Stations (AS) and Ground Stations (GS), addressing the limitations of traditional communication links such as the VHF Digital Link Mode 2 (VDLM2) [11], [17]. LDACS is designed to enhance flight safety, security, and regularity in the aviation industry.

The International Civil Aviation Organization (ICAO) is actively pushing for the standardization of LDACS to facilitate its global adoption [18]. The LDACS architecture includes a Ground Station Controller (GSC) that connects the LDACS network to the broader Air Traffic Network (ATN). This integration supports both Air Traffic Services (ATS) and Aeronautical Operational Control (AOC) end systems, which are crucial for maintaining safe and efficient flight operations [19]. Each GS has the capability to communicate with up to 512 AS simultaneously. The GS communicates with the AS via the Forward Link (FL). Conversely, the AS communicates with the GS through the Reverse Link (RL).

The LDACS frame structure plays a crucial role in its operation. It is based on a Super Frame (SF) that starts with a Broadcast slot in the FL direction. The SF is further divided

into four Multi Frames (MFs), and each MF consists of nine Orthogonal Frequency-Division Multiplexing (OFDM) frames. These OFDM frames contain three Forward Link Physical Layer Service Data Units (PHY-SDUs), which can transmit either user or Common Control data. This functionality enables the GS to allocate resources to an AS, facilitating efficient communication and resource management within the LDACS network [20], [21].

On RL, the Super Frame begins with a Random Access slot, allowing an AS to request access to an LDACS cell. Each Super Frame on this link is comprised of four Multi Frames, and each Multi Frame consists of 162 Reverse Link PHY-SDUs, also known as Orthogonal Frequency-Division Multiple Access (OFDMA) tiles. These tiles are responsible for transmitting Dedicated Control data, which enables an AS to request resource allocation for sending data on the Reverse Link and transmitting user data. This design ensures efficient communication and resource allocation within the LDACS network [22]. Overall, the LDACS frame structure, with its Super Frame, Multi Frame, and PHY-SDU organization, enables effective data transmission, resource allocation, and control mechanisms within the LDACS network.

B. Related Work

Several studies have been conducted to evaluate and improve the security of LDACS. Mäurer et al. [23] undertook a comprehensive assessment of the cybersecurity risks associated with LDACS. Their research identified potential areas and requirements for enhancing the security of LDACS, providing crucial insights into strengthening its cybersecurity measures. In another study, Mäurer et al. [12] introduced an improved architecture and an algorithmic framework for LDACS security. They focused on crucial layers within the LDACS protocol stacks and implemented essential security measures, including end-to-end encryption, message integrity, authenticated key agreement, and entity authentication. Additionally, they employed a Public Key Infrastructure (PKI) approach to secure LDACS communication.

Building on this, the authors in [24] proposed an identity-based variation of the Station to Station (STS) protocol, exploring different types of Diffie-Hellman Key Exchange (DHKE). The proposed protocol aimed to enhance the authentication and key agreement process within LDACS. In a more recent study, authors in [14] proposed securing LDACS using PUF, Challenge-Response (CR), and DHKE protocols. They proposed a PUF-based mutual authentication key exchange scheme that enhances LDACS's cybersecurity architecture. Lastly, in [13], the authors proposed an instantiated version of the ISO/IEC 11770:2021 Key Agreement Mechanism 7 for LDACS, with Elliptic Curve Diffie-Hellman and Elliptic Curve signatures for pre-quantum, and with CRYSTALS-Kyber and FALCON signatures for post-quantum security. Despite ongoing efforts to enhance LDACS security, one important issue remains: LDACS does not address the important goal of aircraft anonymity. This drawback highlights the need for further research and development in LDACS security.

The existing literature provides a valuable foundation for our work. Building upon prior studies, we aim to delve deeper into the subject matter. Specifically, we aim to identify effective countermeasures, optimize security protocols, evaluate alternative security methodologies, and enhance the authentication processes within LDACS. In this paper, we address the following research questions:

- 1) How can we develop an improved mutual authentication process within LDACS that not only enhances security and reduces communication costs, but also ensures the anonymity of AS during communications with GSC?
- 2) How can a secure session key be established between the AS and the GSC in LDACS to ensure the confidentiality and integrity of communications while reducing computational overhead?
- 3) How can we protect LDACS from post-quantum cyber attacks?

III. ADVERSARY MODEL AND SECURITY GOALS

In the LDACS network, a GSC controls several GS, and a single GS handles up to 512 AS. To maintain security and trust within this network, all entities need to mutually authenticate with each other. To ensure the security of the proposed scheme, it is built upon four fundamental security assumptions [25].

- **Eavesdropping capability:** The attacker is assumed to intercept and listen to all messages exchanged between AS and GS. This means the attacker can passively gather information by simply observing the communication.
- **Message interception and alteration:** The attacker can not only intercept the communication but also modify the messages sent between the aircraft and the ground station. This could result in harmful consequences like issuing false commands or altering critical information.
- **Impersonation:** In this scenario, the attacker impersonates a legitimate GSC or AS to send misleading or harmful messages. This typically involves spoofing the identity of the legitimate entity and requires prior knowledge of system protocols and potential security weaknesses.
- **Replay attacks:** If an attacker executes a replay attack within LDACS, they could intercept and resend valid messages. The system, deceived into acknowledging these resubmissions as original transmissions, may execute unauthorized commands, compromising security and potentially risking flight safety. Such an attack could disrupt operational harmony and cause the system to accept inaccurate information as valid.

Bilzhause et al. [26] initially outlined five objectives to secure LDACS, which were later expanded to eight objectives [15]. However, in our study, we concentrate on the central security objective of LDACS, which includes ensuring authenticity, performing integrity checks, and maintaining confidentiality. Moreover, we aim to protect anonymity and secure LDACS against post-quantum cyber attacks [27].

- **Mutual Authentication:** Mutual authentication is important in LDACS because it establishes a trusted and

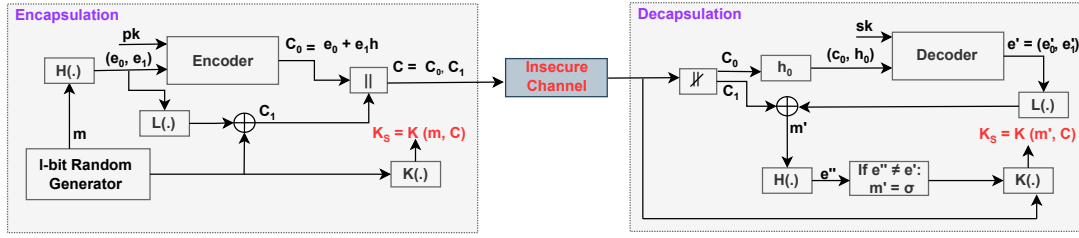


Figure 1: Working principle of the BIKE protocol [16].

secure communication between the AS and GSC. It protects against unauthorized access and attacks, ensuring confidentiality and integrity of data. It builds trust and reliability, supporting efficient air-ground communication in the aviation industry.

- **Integrity:** Integrity is important in LDACS to ensure that transmitted data remains accurate, unaltered, and trustworthy. It prevents unauthorized modifications, ensures system reliability, builds trust, and protects against malicious attacks.
- **Confidentiality:** In LDACS, confidentiality prevents unauthorized disclosure of sensitive information transmitted between aircraft and ground systems. It necessitates using encryption techniques, secure key management, access control mechanisms, and encoding to deter unauthorized access and eavesdropping. The goal is to ensure that only authorized entities can interpret the transmitted data, thereby maintaining the privacy of information within the LDACS system.

LDACS endeavors to establish a comprehensive and highly secure communication system for aviation communication networks by prioritizing maintaining confidentiality, integrity, and authenticity of exchanged information.

IV. PRELIMINARIES

In this section, we explored the preliminary requirements for our proposed solution. We extensively used two crucial components: PUFs and the BIKE protocol.

A. Physically Unclonable Function (PUF)

A PUF is a crucial component for securing communications in cryptography. We utilize PUFs in our system for their unique property of transforming given challenges (C) into corresponding responses (R), represented as $R = PUF(C)$. Each PUF within a circuit manifests a unique challenge-response pair rooted in the system's physical structure. This uniqueness makes PUF outputs unpredictable and challenging to replicate, providing robust authenticity to the system. The complexity of creating PUFs and their resistance to cloning or manipulation make them an ideal solution for securing communications in the LDACS system. Any modifications to the system directly affect the PUF outputs, rendering them nearly impossible to duplicate, which significantly enhances security against malicious activities [28]–[30].

B. Bit Flipping Key Encapsulation (BIKE)

The BIKE cryptographic scheme, which is currently under consideration for standardization by the National Institute of Standards and Technology (NIST) [31], employs the Quasi-Cyclic Moderate Density Parity Check (QC-MDPC) public key encryption to ensure secure communication. The scheme comprises three components: key generation, encapsulation, and decapsulation. In the scheme, Bob first generates both public and private keys, then shares the public key with Alice. Alice then calculates the ciphertext ($C = (C_0, C_1)$) and a symmetric key K_S . Here, the QC-MDPC scheme determines C_0 , and C_1 acts as a guard against possible manipulations. Upon receiving the ciphertext through an insecure channel, Bob decodes C_0 using the QC-MDPC decoder (BGF) and confirms its integrity using C_1 [16]. Following this, Bob derives the symmetric key K_S that matches Alice's key. The circulant matrix blocks within the process are interpreted as polynomial rings to optimize computations. For example, an r -bit sequence is represented as a polynomial $(a_0 + a_1x + a_2x^2 + \dots + a_{r-1}x^{r-1})$ that belongs to $R = F_2[x]/(x^r - 1)$, where F_2 represents a finite field comprising two elements. The functioning of the BIKE protocol is illustrated in Fig. 1.

1) **Key Generation: Private Key:** The private key in BIKE is derived from the QC-MDPC variant and consists of three components: the parity-check matrix of the (r, n, w) -QC-MDPC code with $n_0 = 2$ circulant blocks, random polynomials h_0 and h_1 with Hamming weight $|h_0| = |h_1| = w/2$, and a randomly generated sequence σ of length l . The private key is $sk = (h_0, h_1, \sigma)$.

Public Key: The public key in BIKE is derived from the private key components h_0 and h_1 . To obtain the public key h , the inverse of h_0 (denoted as h_0^{-1}) is computed, and then h is formed by multiplying h_1 with h_0^{-1} . Therefore, the public key is represented as $pk = h = h_1 * h_0^{-1}$.

2) **Encapsulation:** In the BIKE cryptographic scheme, encapsulation uses the public key ' h ' to generate the ciphertext ' C ' and the symmetric key ' K_S ', involving three random oracles (modeled as hash functions): $H : \{0, 1\}^l \rightarrow \{0, 1\}^{2^r}$, $L : \{0, 1\}^{(2^r)} \rightarrow \{0, 1\}^l$, and $K : \{0, 1\}^{(2^r+l)} \rightarrow \{0, 1\}^l$. An l -bit vector ' m ' is selected, then $H(m)$ is used to compute error vectors (e_0, e_1) , and C is calculated as $(C_0, C_1) = (e_0 + e_1.h, m \oplus L(e_0, e_1))$. Finally, ' K_S ' is computed as $K(m, C)$.

3) **Decapsulation:** In decapsulation, the private key ' SK ' and ciphertext ' C ' are input to generate ' KS '. Syndrome ' S '

Table I: Notation and Description

Notation	Description	Notation	Description
International Civil Aviation Organization Address	$ICAO_A$	Aeronautical Fixed Telecommunication Network	$AFTN_A$
Private and Public element of AS	S_{EAS}, P_{EAS}	Message, where i_{th} is the message number	M_i
Temporary (pseudo) identity of AS	τ_{AS}	Addition, Subtraction, Bit-wise XOR	$+, -, \oplus$
Operator: Equality, Not Equal	$\equiv?, \neq$	Hash, Concatenation, Scalar Multiplication	$h, , \cdot$
Truncated to first 24 bits	T_{24}	Variables	$\alpha, \beta, \gamma, \theta, \omega, j, \sigma$
Symmetric Key, Syndrome, Message Authentication Code	$K_s, S, (h_\theta, h_\beta)$	If true then continue else abort	$T - C/E - A$
Authentication Successful, Connection Termination	AS, CT	Nonces	N_1, N_2, N_3, N_4
Random Oracles (Hash Functions)	H, L, K	Ciphertext, Error Vectors,	C, e_0, e_1
Challenge, Response, Embedded PUF	C_{AS}, R_{AS}, P_{AS}	Fetch From Memory	$Fetch_M$
MAKE	Mutual Authentication & Key Exchange		

is computed as $S = C_0 \cdot h_0$ and applied to the e Black-Grey-Flip (BGF) decoder to obtain error vectors e_0' and e_1' . Next, m' is calculated using $m' = C_1 \oplus L(e_0', e_1')$ and if $H(m') \neq (e_0', e_1')$, m' is set to σ else $K_S = K(m', C)$ is computed.

V. PROPOSED MAKE PROTOCOL

This section presents a proposed method for mutual authentication and establishing a secret session key to secure air-ground communication between AS and GSC in the LDACS network. The symbols and terms used to describe the protocol's operation are provided in Table I.

In the LDACS system operation, several assumptions are made: (1) The GSC is considered a tamper-prone entity. (2) Public and secret elements are generated using the principle of the BIKE protocol. (3) The GSs are already authenticated by the GSCs. (4) PUF has been securely embedded into the aircraft's hardware. (5) The hashed information cannot be retrieved. (6) Both the aircraft and the GSC can perform cryptographic operations and possess the same infrastructural knowledge (i.e., random oracles (H, L, K)). (7) The Pseudorandom Number Generator (PRNG) behaves similarly to a True Random Number Generator (TRNG).

A. Registration Phase

The registration phase is conducted offline before the LDACS radio is put into actual service. This critical phase is carried out in a secure, tempest-proof environment by verified employees, providing a trusted setting that protects against cyber-attacks. Given these protective measures, the registration phase is assumed to be secure, thus ensuring the integrity and security of the processes involved. The complete registration process is illustrated in Fig. 2 and explained as follows:

- **Step 1:** The process begins when the GSC generates a challenge (C_{AS}) for the AS. The GSC then formulates a message which includes this challenge (C_{AS}) along with the GSC's $AFTN_A$ address. Subsequently, the GSC sends this message, denoted as M_1 , to the AS.
- **Step 2:** Upon receiving M_1 , the AS proceeds to calculate the response R_{AS} for the challenge C_{AS} , denoted as $R_{AS} = P_{AS}(C_{AS})$. This response is then hashed, resulting in $\theta = h(R_{AS})$, which will be used in subsequent steps. The AS retrieves the values P_{EAS} and

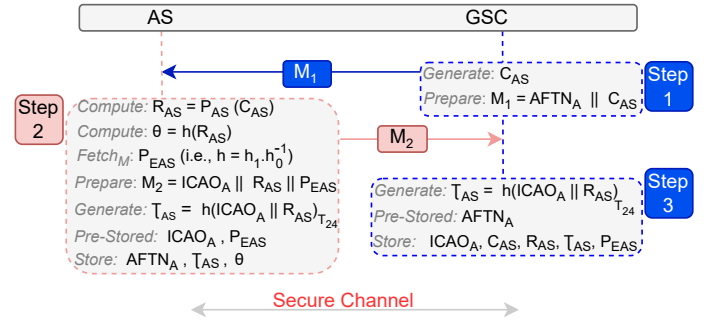


Figure 2: Aircraft Station Registration Phase.

$ICAO_A$ from its memory and prepares the message $M_2 = ICAO_A || R_{AS} || P_{EAS}$. After preparing the M_2 , the AS sends this M_2 to the GSC for further processing. Simultaneously, the AS generates its temporary identity τ_{AS} by hashing the concatenation of $ICAO_A$ and R_{AS} and truncated to first 24 bits. Lastly, the AS securely stores the values $AFTN_A$, τ_{AS} , and θ for future need.

- **Step 3:** The GSC receives the message M_2 and generates a temporary identity τ_{AS} for the AS by taking the hash of the concatenated values of $ICAO_A$ and R_{AS} . The τ_{AS} is then truncated to the first 24 bits to keep the original $ICAO_A$ length. Lastly, GSC stores all the important computed and received values, e.g., $ICAO_A$, C_{AS} , R_{AS} , τ_{AS} , and P_{EAS} .

B. Mutual Authentication

This phase is initiated when the AS requires information exchange with the GSC. Before any secret key negotiation between the AS and the GSC, it is essential to validate the legitimacy of both the AS and the GSC. Fig. 3 provides a comprehensive overview of the authentication process, which is further discussed below:

- **Step 1:** The AS initiates the process by generating a nonce, N_1 , for freshness and retrieves the temporary identity, τ_{AS} , from its memory. To prove data integrity and authenticity, the AS calculates a MAC, $\alpha = h_\theta(N_1 || \tau_{AS})$. Subsequently, the AS creates the

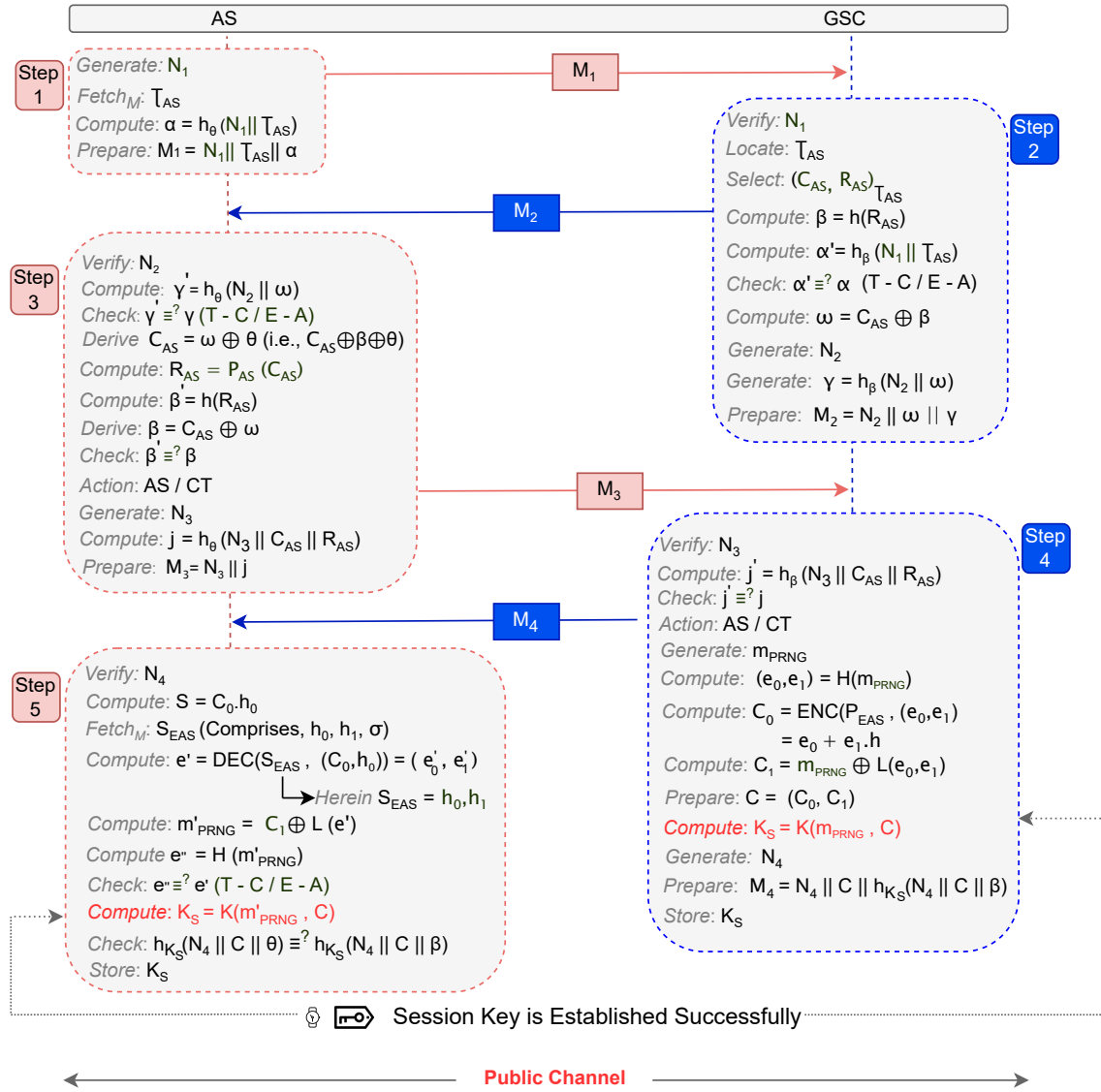


Figure 3: Mutual authentication and key establishment process.

message $M_1 = N_1 || \tau_{AS} || \alpha$ and sends it to the GSC for processing without revealing sensitive data such as the aircraft's real identity.

- **Step 2:** Upon receiving message M_1 , the GSC initially verifies the freshness of nonce N_1 . If the nonce is not fresh, the connection is immediately terminated. If it is fresh, the GSC retrieves the temporary identity, τ_{AS} , from its memory and selects the corresponding challenge-response pair, C_{AS} and R_{AS} . For authentication and data integrity, the GSC computes the MAC as $\alpha' = h_{\beta}(N_1 || \tau_{AS})$ and compares this with the received α . Given θ/β is known only to the legitimate AS and GSC, this should validate the authenticity. If it does not, the connection is terminated instantly. To authenticate itself to the AS, the GSC constructs ω through a bitwise XOR operation of C_{AS} and β . The GSC then generates a new nonce, N_2 , and computes a MAC as $\gamma = h_{\beta}(N_2 || \omega)$ to

prove freshness, authenticity, and data integrity at AS. Lastly, the GSC sends message M_2 , containing N_2 , ω , and γ , to the AS without revealing any information about C_{AS} and R_{AS} .

- **Step 3:** Upon receiving M_2 , the AS checks the freshness of nonce N_2 and calculates the MAC as $\gamma' = h_{\theta}(N_2 || \omega)$. The AS then confirms the integrity and authenticity of the received information by ensuring $\gamma' \equiv \gamma$. If the information is authenticated and deemed fresh, the AS proceeds to derive the challenge $C_{AS} = \omega \oplus \theta$, {i.e., $C_{AS} \oplus \beta \oplus \theta = C_{AS}$ }. Recalling the registration phase, the AS retrieves the hash of its response R_{AS} , stored in variable θ . The retrieved challenge is input into the embedded PUF, $P_{AS}(C_{AS})$, to generate the corresponding response, R_{AS} . Once the response is available, the AS computes the message digest $\beta' = h(R_{AS})$, deduces β by calculating $\omega \oplus C_{AS}$, and compares $\beta' \equiv \beta$ to authenticate the

GSC. This check must hold true since only a legitimate GSC would know the response to the challenge. If it fails, the connection is immediately terminated. In order to authenticate itself, the AS generates a fresh nonce, N_3 , computes the MAC as $j = h_\theta(N_3||C_{AS})||R_{AS}$, and prepares the message $M_3 = N_3||j$ to send to the GSC. This procedure is performed without revealing sensitive C_{AS} and R_{AS} information, thus preventing any potential alterations or replay attacks.

- **Step 4:** Upon receiving M_3 , the GSC performs several verification checks to verify the integrity and authenticity of the M_3 . First, it validates the freshness of nonce N_3 to ensure the message is neither replayed nor outdated. Next, the GSC calculates the MAC as $j' = h_\beta(N_3||C_{AS})||R_{AS}$ to ensure the integrity and authenticity of the message and then compares the received value, i.e., $j' \stackrel{?}{=} j$. Since only the legitimate AS can generate the correct R_{AS} , equality in this comparison signifies the successful authentication of AS at GSC.

The next phase involves establishing a secret session key between the AS and GSC inspired by the BIKE protocol. The GSC generates an '1' bit random vector m using a PRNG. Based on the BIKE protocol, GSC uses three random oracles (modeled as hash functions), denoted as H , L , and K , which are modeled as random oracles. The GSC inputs m_{PRNG} to the first random oracle to produce two error vectors, $(e_0, e_1) = H(m_{PRNG})$. The GSC uses the error vectors e_0 and e_1 and the public element P_{EAS} of the AS to produce the first ciphertext component, $C_0 = e_0 + e_1 \cdot h$. The second ciphertext component, $C_1 = m_{PRNG} \oplus L(e_0, e_1)$, helps the recipient AS to verify the integrity of C_0 . Finally, GSC inputs the information m_{PRNG} and C to its third random oracle, K . The final random oracle is critical as it considers the information m_{PRNG} and ciphertext C to produce the secret session key K_S . Lastly, GSC computes the MAC of nonce N_4 , C , and β using key K_S and prepares the message $M_4 = N_4||C||h_{K_S}(N_4||C||\beta)$ and sends to AS for further processing. The expression $h_{K_S}(N_4||C||\beta)$ enables the AS to verify the establishment of the same key on both sides. Lastly, GSC stores the key K_S .

- **Step 5** Upon receiving message M_4 , the AS first verify the freshness of nonce N_4 . It then computes the syndrome through the expression $S = C_0 \cdot h_0$. It's important to note that an attacker cannot generate the syndrome, as it requires knowledge of h_0 , which is only known to the AS. Next, the AS uses its secret element $S_{EAS} = (h_0, h_1, \sigma)$ and syndrome to deduce the error vectors using a BGF decoder, $e' = e'_0, e'_1 = DEC(S_{EAS}, (C_0, h_0))$. The AS then evaluates the integrity of the received ciphertext by computing $m'_{PRNG} = C_1 \oplus L(e')$, $e'' = H(m'_{PRNG})$ and comparing them: $e'' \stackrel{?}{=} e'$. If $e'' \stackrel{?}{=} e'$ are equal, it indicates that the received error vectors e_0, e_1 and the random vector m'_{PRNG} have been correctly recovered. In case of inequality, implying a mismatch or alteration, the connection is terminated to prevent

further communication with an untrusted entity. Finally, the AS computes the secret session key by calculating $K_S = K(m'_{PRNG}, C)$. The AS then computes the MAC using the freshly generated key $h_{K_S}(N_4||C||\theta)$ and compares it with the received information from the GSC, $h_{K_S}(N_4||C||\theta) \stackrel{?}{=} h_{K_S}(N_4||C||\beta)$. Equality indicates that the AS has generated a similar key, whereas inequality necessitates the termination of the connection. The AS then stores the key K_S .

VI. INFORMAL SECURITY ANALYSIS

In this section, we performed an informal security analysis of our proposed scheme.

Theorem 1. Ensuring Mutual Authentication.

Proof. The proposed scheme allows only legitimate parties, such as the AS and GSC, to participate in the communication. Our proposed scheme uses PUF to verify the authenticity of the entities involved in the communication. For example, if the AS receives the message $M_2 = N_2||\omega||\gamma$, the AS generates a challenge $C_{AS} = \omega \oplus \theta$, which is used by the PUF to produce a response, $R_{AS} = P_{AS}(C_{AS})$. Once the response is ready, the AS computes the message digest $\beta' = h(R_{AS})$, deduces β by calculating $\omega \oplus C_{AS}$, and compares $\beta' \stackrel{?}{=} \beta$ to authenticate the GSC. The check ensures only a legitimate GSC knows the challenge response. If it fails, the connection is terminated. Subsequently, the AS prepares a message $M_3 = N_3||j$ for the GSC, where $j = h_\theta(N_3||C_{AS})||R_{AS}$ is the MAC. Upon receipt, the GSC verifies the message and the AS's authenticity by ensuring $j' \stackrel{?}{=} j$. Since only the legitimate AS can generate the correct R_{AS} , equality in this comparison signifies the successful authentication of AS at GSC. $\square$

Theorem 2. Preserving Message Integrity.

Proof. The proposed scheme utilizes a MAC to prevent message modification. Consider a scenario where an attacker attempts to modify the nonce N_1 or α in the message $M_1 = N_1||\tau_{AS}||\alpha$. Upon receiving M_1 , the GSC computes the MAC as $\alpha' = h_\beta(N_1||\tau_{AS})$ and compares it with the received α to verify integrity. The legitimacy and integrity are ensured using an undisclosed parameter, denoted as θ or β , known only to the authentic AS and GSC. The connection is immediately terminated if the comparison between θ/β fails. $\square$

Theorem 3. Assuring Confidentiality.

Proof. The proposed scheme ensures the confidentiality of sensitive information, such as the aircraft's identity (τ_{AS}), PUF challenges (C_{AS}), response (R_{AS}), and session keys (K_S). Even if an attacker captures the message $M_2 = N_2||\omega||\gamma$, the collision-resistant property of hash functions and the unavailability of the (C_{AS}) prevent the attacker from retrieving any sensitive information. Similarly, in message M_3 , C_{AS} and R_{AS} are sent as a message digest, maintaining confidentiality. In message M_4 , the generated K_S is used as a key for the MAC but is not included in the message. Additionally, M_1 also

ensures the confidentiality of sensitive information. As a result, none of the messages disclose any confidential information. $\square$

Theorem 4. Establishing Session Key Agreement.

Proof. The GSC and AS establish the session key independently using the BIKE protocol. The GSC generates a random vector m and computes error vectors $(e_0, e_1) = H(m_{PRNG})$ using a random oracle H . The GSC then computes ciphertext components $C_0 = e_0 + e_1 \cdot h$ and $C_1 = m_{PRNG} \oplus L(e_0, e_1)$. Upon receiving the GSC's message, the AS computes the syndrome $S = C_0 \cdot h_0$. The AS recovers the error vectors e'_0 and e'_1 using its secret element S_{EAS} and the syndrome S . To ensure the integrity of the received ciphertext, the AS verifies the computed values by comparing $m'_{PRNG} = C_1 \oplus L(e')$ with the received ciphertext C_1 and $e'' = H(m'_{PRNG})$ with the computed error vectors e' . Finally, both the GSC and AS independently compute the session key K_S . The GSC uses the random oracle K with inputs m_{PRNG} and C to derive K_S . Likewise, the AS applies a random oracle K to the recovered m'_{PRNG} and the received ciphertext C to obtain K_S . This process ensures the establishment of a shared session key, enabling secure communication between the GSC and AS. $\square$

Theorem 5. Ensuring Message Freshness.

Proof. The proposed approach is resistant to replay attacks. Let's consider a scenario where an intruder captures the message $M_1 = N_1 || \tau_{AS} || \alpha$. If the intruder attempts to impersonate a legitimate aircraft by relaying this message (M_1) to the GSC, the system detects it as the message contains an old random nonce. As a result, the GSC halts the process and immediately terminates the session. Similarly, the other messages, namely M_2 , M_3 , and M_4 , are also protected against replay attacks. $\square$

Theorem 6. Providing Anonymity.

Proof. The proposed protocol guarantees both identity and location anonymity. During the registration phase, the AS generates a temporary identity, denoted as $\tau_{AS} = h(ICA_{OA} || R_{AS})$, using a message digest. This τ_{AS} will be used in all future communications to replace the actual identity. This strategy ensures that any exchanged messages, such as $M_1 = h(N_1 || \tau_{AS} || \alpha)$, do not disclose any sensitive identity-related information. Furthermore, the protocol protects location anonymity by inhibiting the release of the aircraft's exact geographical coordinates in the messages exchanged. This level of anonymity is maintained in all subsequent messages, including M_2 , M_3 , and M_4 . $\square$

VII. PERFORMANCE AND COMPARATIVE ANALYSIS

The findings presented in Table II demonstrate that the proposed protocol has successfully achieved a comprehensive set of security properties, including data privacy, message integrity, freshness, anonymity, untraceability, and mutual authentication. Incorporating these security properties significantly reinforces the proposed security protocol, enabling it

Table II: Comparison of Security Features: Proposed Protocol vs. Conventional Protocols.

Goals		[14]	[24]	Proposed
Protection Against	Replay	✓	✓	✓
	DoS	*	*	*
	Impersonation	✓	✓	✓
	Modification	✓	✓	✓
	MITM	✓	✓	✓
	Cloning	✓	✓	✓
	Known Key	✓	✓	✓
Post Quantum Security		*	*	✓
Message Integrity		✓	✓	✓
Confidentiality		✓	✓	✓
Message Freshness		✓	✓	✓
Anonymity		×	×	✓
Security Analysis		*	✓	✓
Mutual Authentication and Key Exchange		✓	✓	✓

✓: goal accomplished, ×: goal unaccomplished, *: To some extent

to effectively withstand various attacks, such as replay, impersonation, modification, MITM, cloning, etc. By comparing the results in Table II, it becomes apparent that the existing schemes [13], [14], [24] may have certain areas where they could be further improved in terms of offering comprehensive defense against all prominent attacks. Most importantly, these schemes do not provide identity concealment.

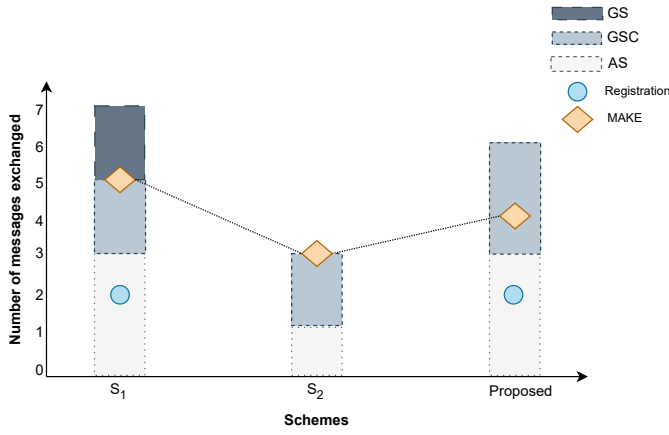
Table III presents a comparison between the computational costs associated with our proposed protocol and conventional methods during the registration, mutual authentication, and key agreement phases. We thoroughly analyze and discuss the communication expenses based on the existing literature [14] [24]. Compared with the schemes detailed in [14] and [24], our proposed protocol demonstrates a comparable computational cost during the registration, authentication, and key agreement phases. However, as indicated in Table II, our proposed scheme offers additional provisions for anonymity and assured protection against post-quantum attacks.

Figure 4 depicts the number of messages exchanged during the registration and mutual authentication phases. Since the schemes referenced in [24] are certificate-based, they don't require a registration phase. In line with the current literature in [14], our proposed model exchanges two messages during the registration phase, matching the standards set by Mäurer et al. [14]. However, a divergence occurs in the mutual authentication and key agreement phase. Our scheme successfully reduces the number of exchanged messages to 4. In contrast, the existing solutions by Mäurer et al. in [14] require 5, and in [24] requires 3 messages, respectively, during the same mutual authentication and key agreement phase. In conclusion, the key advantage of our proposed protocol lies in its efficiency during the mutual authentication and key agreement phase, when compared to existing PUF-based protocols for LDACS. This improvement reduces the communication overhead, thus enhancing the overall performance and making our protocol a more viable choice for secure and efficient communication

Table III: Comparison of Computational Costs: Proposed Scheme vs. Traditional Schemes.

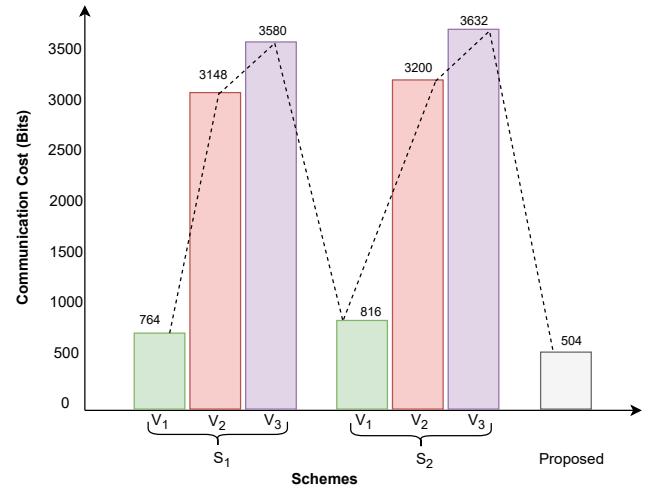
Scheme	Phase	Aircraft Station	Ground Station Controller	Total Cost
[14]	Registration	PUF_{1T}	-	PUF_{1T}
	MAKE	$MAC_{4T} + XOR_{6T} + SE_{1T} + PUF_{2T} + R_{1T} + HKDF_{1T}$	$MAC_{4T} + XOR_{5T} + SD_{1T} + R_{1T} + HKDF_{1T}$	$MAC_{8T} + XOR_{11T} + SE_{1T} + SD_{1T} + PUF_{2T} + R_{2T} + HKDF_{2T}$
[24]	Registration	Not Disclosed	Not Disclosed	-
	MAKE	$AE_{1T} + AD_{1T} + R_{1T} + H_{2T} + HKDF_{1T} + EXP_{2T} + Mod_{2T}$	$AE_{1T} + AD_{1T} + R_{1T} + H_{2T} + HKDF_{1T} + EXP_{2T} + Mod_{2T}$	$AE_{2T} + AD_{2T} + R_{2T} + H_{4T} + HKDF_{2T} + EXP_{4T} + Mod_{4T}$
Proposed	Registration	$PUF_{1T} + H_{2T}$	H_{1T}	$PUF_{1T} + H_{3T}$
	MAKE	$D_{1T} + MAC_{4T} + XOR_{3T} + PUF_{1T} + R_{2T} + SM_{1T} + H_{4T}$	$E_{1T} + MAC_{4T} + XOR_{2T} + R_{3T} + H_{4T}$	$D_{1T} + E_{1T} + MAC_{8T} + XOR_{5T} + PUF_{1T} + R_{5T} + SM_{1T} + H_{8T}$

Acronyms: *PUF*: Physically Unclonable Function, *MAC*: Message Authentication Code, *H*: Hash, *XOR*: Bit-wise XOR, *SE*: Symmetric encryption, *SD*: Symmetric decryption, *AE*: Asymmetric encryption, *AD*: Asymmetric decryption, *R*: Random number, *HKDF*: Hash-Based Key Derivation Function, *EXP*: Exponentiation, *MOD*: Modulus, *E*: Encoder, *D*: Decoder, *SM*: Scalar Multiplication,


 Figure 4: Comparison of communication cost: Proposed Scheme vs. Traditional Schemes. Schemes in comparison, S_1 : [14], S_2 : [24].

within the LDACS system, particularly when PUF schemes are desired.

Fig. 5 compares the number of bits exchanged by the AS in the proposed solution versus traditional solutions. We focus specifically on AS as they typically have fewer resources than GSC. Both [14] and [24] have three variants for mutual authentication and key exchange, namely V_1 : *ECDH*, V_2 : *SIDH*, and V_3 : *DHKE*. For [14], V_1, V_2, V_3 require 764, 3148, and 3580 bits, respectively. Similarly, [24] requires 816, 3200, and 3632 bits for V_1, V_2 , and V_3 , respectively. In contrast, our proposed scheme requires only 504 bits on the AS side for mutual authentication and key exchange. In our proposed scheme, the message M_1 comprises 264 bits, while M_2 comprises 240 bits. The components of M_1 include a 48-bit header, a 24-bit identity, a 64-bit nonce, and a 128-bit MAC tag. On the other hand, M_2 comprises a 48-bit header, a 64-bit nonce, and a 128-bit MAC tag. Combining both M_1 and M_2 , the total AS message size is 504 bits. This significantly reduces overhead and latency, making our solution more efficient.


 Figure 5: Comparison of communication cost (bits): Proposed Scheme vs. Traditional Schemes. Schemes in comparison, S_1 : [14], S_2 : [24].

Our solution outperforms traditional protocols by reducing the number of bits exchanged, improving efficiency and resource utilization.

VIII. CONCLUSIONS AND FUTURE WORK

LDACS, a significant advancement within the FCI, offers innovative solutions to address existing aviation communication limitations with its improved bandwidth, capacity, and coverage over traditional systems. Moreover, its robust security features ensure the confidentiality, integrity, and availability of aeronautical communications. While encryption mechanisms and authentication protocols are incorporated to protect against unauthorized access and cyber threats, the standardization process for mutual authentication and key establishment protocols is still ongoing and warranting continued attention for optimal security measures. Our proposed solution provides a robust, secure, lightweight protocol designed explicitly for

air-ground LDACS communication. Leveraging lightweight cryptographic primitives such as MAC, nonce, PUF, and bitwise XOR operations, this protocol ensures secure, efficient communication between aircraft and ground systems. Further, integrating the BIKE protocol enables robust, secure session key generation, ensuring resilience against post-quantum cyberattacks and safeguarding LDACS's long-term security.

To assess our proposed scheme, we compared its security features, computational costs, and communication overhead against existing LDACS security protocols. The results clearly demonstrate that our protocol enhances security by incorporating identity and location anonymity, while also providing resilience against post-quantum cyber-attacks. Furthermore, our approach demonstrated a low message exchange overhead, requiring only four messages to complete the mutual authentication phase between the AS and GSC. Since AS has fewer resources than GSC, we calculated the communication cost specifically for AS. The results showed it only necessitates 504 bits, thereby reducing communication latency and emphasizing our protocol's efficiency in securing LDACS systems. Our future research will focus on securing the handovers within LDACS, encompassing both intra-ATN and inter-ATN handovers.

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